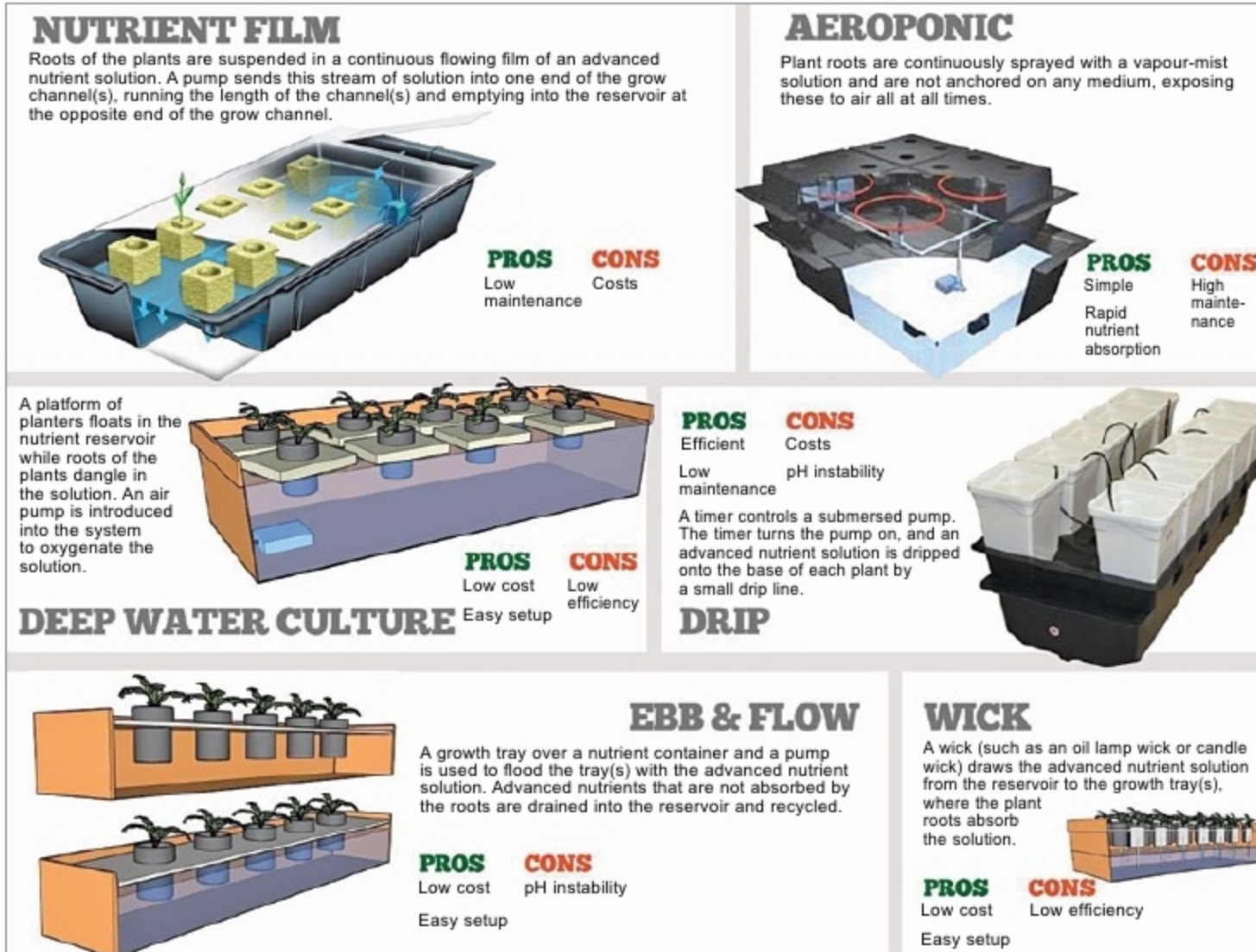




Types of hydroponic systems (Credit: <http://hydroponicsfarm.co.in>)

crops. Watering cycles, auto-refill levels and pH levels are monitored and maintained using a microcontroller.

This method of growing plants without soil uses mineral-nutrient solutions in a water solvent. Plants are grown with only their roots dipped in the solution and supported by perlite or gravel. Nutrient-rich water is recirculated that provides high water consumption efficiency. The vertical design allows higher density of plants in a much smaller footprint. The microcontroller is connected to a Wi-Fi module that senses data through sensors and sends it over a mobile app for plant maintenance.

Innovative hydroponic gardens

Traditional farming uses soil and takes a longer time to decompose. This increases costs and makes the plants prone to diseases. IoT-based vertical gardening provides better results, especially in areas with unsuitable environment and space.

This system has no adverse effect on the quality of crops, soil or envi-

ronment. It allows the control of nutrient values via nutrient solutions. It improves the quality of crops with cost-effective implementations for small farmers in India. The setup helps save land from adverse effects of pesticides and other factors affecting the quality of soil.

There are many innovators who are trying to make vertical gardening easy and productive for you. Team Plug & Plant on Kickstarter creates vertical gardens that make cultivating a wall full of plants and herbs pretty simple.



Gartenzwerg indoor gardening system (Credit: <https://gartenzwerg.me/>)

Plug & Plant is a ready-to-grow soil-free environment with a wall-mounted modular system. Its tank is equipped with temperature, light, water and humidity sensors to gather feedback on what is happening in each plug and communicate via Bluetooth to smartphones.

The entire system is automated, except that you need to add water to the tank. It makes sure each seed receives the right amount of water, once the Android or iOS app knows what kind of seeds are planted. You only need to refill the tank once a month and choose from over 30 varieties of plugs based on how much light these require.

Gartenzwerg is another automated indoor gardening system that enables growing of vegetables, small fruits and herbs using a click of a button, from anywhere in the world. It knows exactly what the plants need and when, because it mimics perfect environmental conditions. Plants grow up to three times faster than any soil-based produce. This system, too, runs on soil-less, water-nutrient solutions.

The plants grow tasty and strong in any season, so you can have fresh strawberries even in winters. You can harvest as and when required and have long-lasting supplies. It lets you know if your plants are thirsty or need attention.

RainMachine takes care of your sprinkler system and helps save water. Using seven-day forecasts, the system uses real-time weather data